Optimization: Critical Points & Lagrange Multipliers

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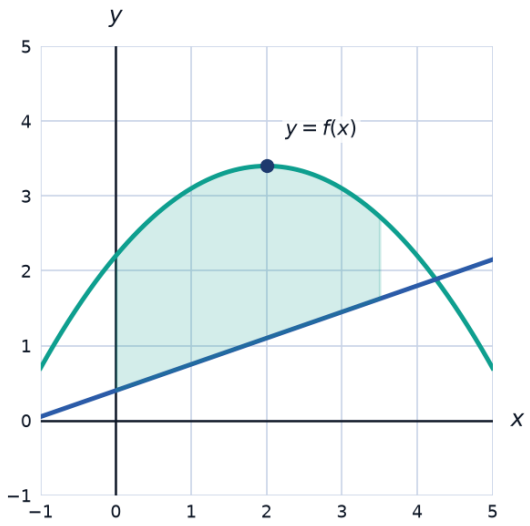
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Finding Critical Points

Find all critical points of the function $f(x, y) = x^3 + y^3 - 3xy$.

2 [Foundational] • Target: The Second Partials Test (Hessian Discriminant)

Consider the surface S which is the portion of the plane $z = 2x + 3y + 1$ lying above the rectangular region $R = [0, 1] \times [0, 2]$ in the xy-plane. Calculate the surface area of S.



3 [Foundational] • Target: Classifying Extrema and Inconclusive Cases

Compute the scalar line integral with respect to arc length $\int_C (x + 2y + 3z)\, ds$, where C is the straight line segment from the origin $(0, 0, 0)$ to $(1, 2, 2)$.

4 [Foundational] • Target: Lagrange Multipliers for Constrained Optimization

Consider the function $f(x, y) = x^3 - 3xy + y^2$. Find the value of $f_{xx}(1, 1)$.

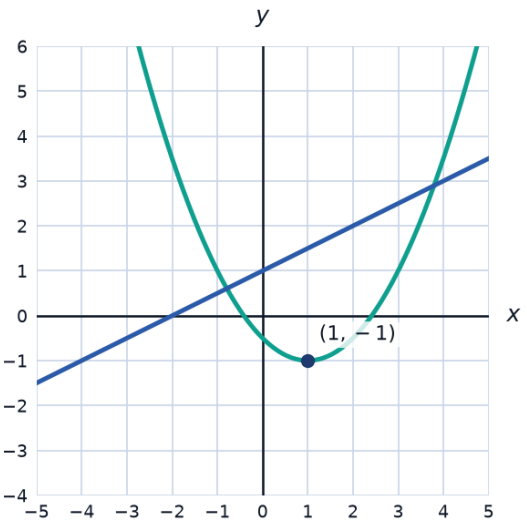
5 [Foundational] • Target: Finding Critical Points

Let the scalar field be given by $f(x, y) = 3x^2 - 5y$. Compute the value of $f_x(2, -3)$.

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6 [Foundational] • Target: The Second Partials Test (Hessian Discriminant)

Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle x^2, xy \rangle$ along the segment from $(3,\ 3)$ to $(1,\ 0)$.



7 [Foundational] • Target: Classifying Extrema and Inconclusive Cases

Find the volume of the solid inside the sphere $\rho = 4$ and above the cone $\phi = \frac{\pi}{4}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Lagrange Multipliers for Constrained Optimization

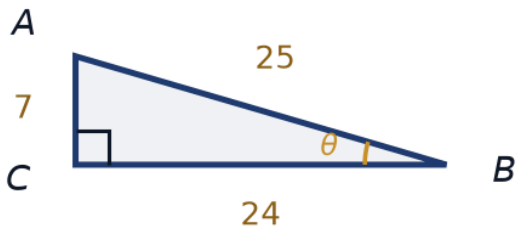
Compute $\iint_D 9(x^2 + y^2)\, dA$ where D is the disk $x^2 + y^2 \le 49$.

9 [Intermediate] • Target: Finding Critical Points

For $f(x, y) = 4x^2y^2 + 3xy$, find the mixed partial $f_{xy}(1,\ -2)$.

10 [Intermediate] • Target: The Second Partials Test (Hessian Discriminant)

Use Green's theorem to evaluate $\oint_C -y^3\, dx + x^3\, dy$, where C is the triangle with vertices $(0,\ 0)$, $(3,\ 0)$, $(0,\ 3)$, oriented counterclockwise.



11 [Intermediate] • Target: Classifying Extrema and Inconclusive Cases

Consider the function $f(x, y) = x^2 + y^2 - 4x - 6y + 13$. Find the coordinates of the critical point for this function.

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12 [Intermediate] • Target: Lagrange Multipliers for Constrained Optimization

Consider the function $f(x, y) = x^3 + y^3 - 3axy$, where a is a positive constant. It is known that $f(x, y)$ has a critical point at (x_0, y_0) where $x_0 = 2$ and $y_0 = 2$. Find the value of a.

13 [Intermediate] • Target: Finding Critical Points

Consider the function $f(x, y) = x^3 - 3xy + y^2$. Find the value of $f_{xx}(1, 1)$.

14 [Intermediate] • Target: The Second Partials Test (Hessian Discriminant)

Consider the function $f(x, y) = x^2 + y^2$. Find the value of $f_{xx}(2, 3)$.

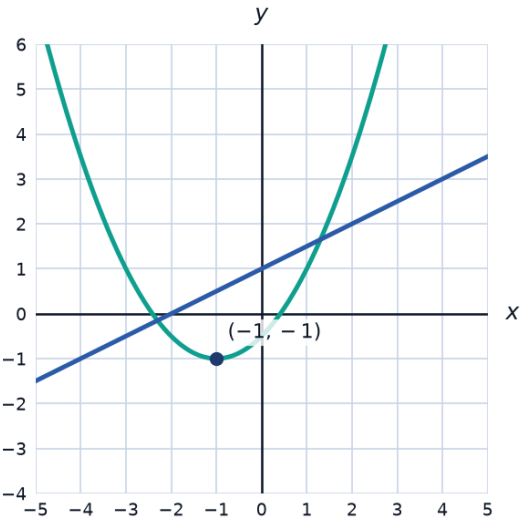
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Classifying Extrema and Inconclusive Cases • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

For $f(x, y) = x^2 - 6xy + 3y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

16 [Challenging] • Target: Lagrange Multipliers for Constrained Optimization • Cross-Lesson: 4.01 Double Integrals

For $f(x, y) = 3x^2 - 6xy + y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).



17 [Challenging] • Target: Finding Critical Points • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

For $f(x, y) = x^2 - 6xy + y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

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18 [Challenging] • Target: The Second Partials Test (Hessian Discriminant) • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

For $f(x, y) = 2x^2 - 6xy + 2y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

19 [Challenging] • Target: Classifying Extrema and Inconclusive Cases • Cross-Lesson: 4.01 Double Integrals

For $f(x, y) = 4x^2 - 6xy + y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

20 [Challenging] • Target: Lagrange Multipliers for Constrained Optimization • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

For $f(x, y) = 3x^2 - 5xy + y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Finding Critical Points

What condition defines a critical point (x_0, y_0) of a differentiable function $f(x, y)$?

a. $\nabla f(x_0, y_0) = \mathbf{0}$, meaning both $f_x(x_0, y_0) = 0$ and $f_y(x_0, y_0) = 0$

b. $f_{xx}(x_0, y_0) = 0$

c. $f(x_0, y_0) = 0$

d. $f_x(x_0, y_0) = f_y(x_0, y_0)$

22 [Foundational] • Target: Finding Critical Points

Find the critical points of $f(x, y) = x^2 + y^2 - 4x + 6y + 10$.

a. $(2, -3)$

b. $(-2, 3)$

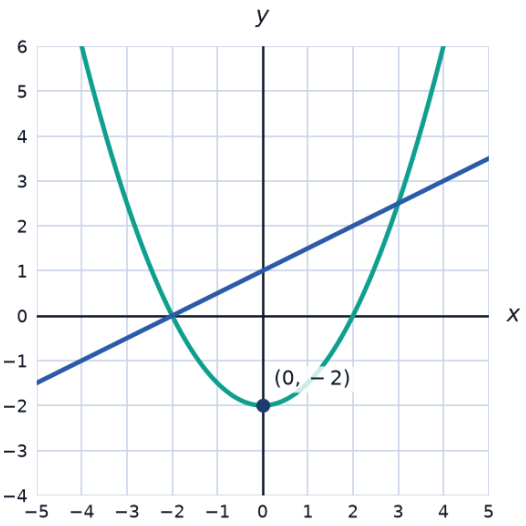
c. $(4, -6)$

d. $(0, 0)$

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23 [Foundational] • Target: Finding Critical Points

Find all critical points of $g(x, y) = x^3 - 3x + y^2$.



a. $(1, 0)$ and $(-1, 0)$

b. $(1, 0)$ only

c. $(0, 0)$

d. $(3, 0)$

24 [Foundational] • Target: The Second Partials Test (Hessian Discriminant)

What is the definition of the Hessian discriminant D in the Second Partials Test?

a. $D = f_{xx}f_{yy} - (f_{xy})^2$

b. $D = f_{xx} + f_{yy}$

c. $D = (f_{xy})^2 - f_{xx}f_{yy}$

d. $D = f_{xx}f_{yy} - f_{xy}$

25 [Foundational] • Target: The Second Partials Test (Hessian Discriminant)

If at a critical point $D > 0$ and $f_{xx} > 0$, what does the test conclude?

a. f has a local minimum at the point

b. f has a local maximum at the point

c. The point is a saddle point

d. The test is inconclusive

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26 [Foundational] • Target: The Second Partials Test (Hessian Discriminant)

If at a critical point $D < 0$, what does the test conclude?

a. f has a saddle point

b. f has a local maximum

c. f has a local minimum

d. The test is inconclusive

27 [Foundational] • Target: Classifying Extrema and Inconclusive Cases

Classify the critical point $(0, 0)$ for $f(x, y) = x^2 - y^2$.

a. Saddle point

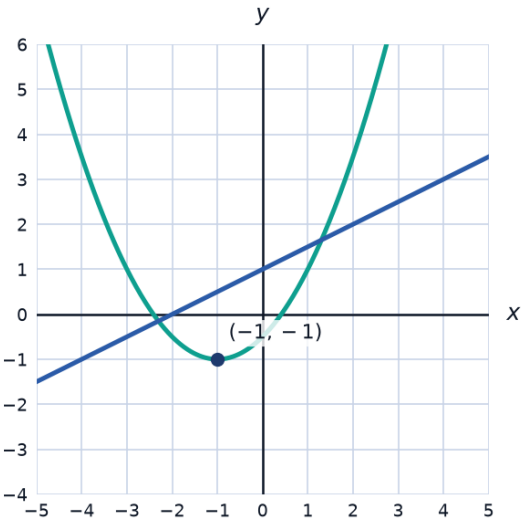
b. Local minimum

c. Local maximum

d. Inconclusive

28 [Foundational] • Target: Classifying Extrema and Inconclusive Cases

What does the Second Partials Test conclude when $D = 0$?



a. The test is inconclusive; higher-order analysis or algebraic examination is needed

b. The point is definitely a saddle point

c. The point is an absolute extremum

d. The function is not differentiable

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Classifying Extrema and Inconclusive Cases

Classify $(2, -3)$ for $f(x, y) = x^2 + y^2 - 4x + 6y + 10$.

a. Local minimum

b. Local maximum

c. Saddle point

d. Inconclusive

30 [Intermediate] • Target: Lagrange Multipliers for Constrained Optimization

What vector equation governs constrained optimization of $f(x, y)$ subject to constraint $g(x, y) = k$?

a. $\nabla f = \lambda \nabla g$ (and $g(x, y) = k$)

b. $\nabla f \cdot \nabla g = 0$

c. $\nabla f + \nabla g = \mathbf{0}$

d. $\nabla f = \mathbf{0}$

31 [Intermediate] • Target: Lagrange Multipliers for Constrained Optimization

Maximize $f(x, y) = xy$ subject to the constraint $x + y = 10$ for $x, y > 0$.

a. Maximum is 25 at $(5, 5)$

b. Maximum is 24 at $(4, 6)$

c. Maximum is 21 at $(3, 7)$

d. Maximum is 16 at $(2, 8)$

32 [Intermediate] • Target: Lagrange Multipliers for Constrained Optimization

What does the Lagrange multiplier λ represent physically/economically in constrained optimization?

a. The sensitivity / shadow price: the rate of change of the optimal value with respect to the constraint constant k ($\frac{df^*}{dk}$)

b. The curvature of the constraint surface

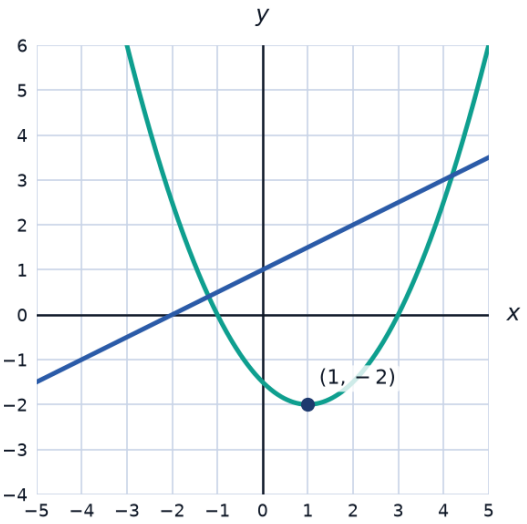
c. The distance from the origin to the optimal point

d. The second derivative along the constraint

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33 **[Intermediate]** • **Target: Finding Critical Points**

For $f(x, y) = 2x^2 - 6xy + 3y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).



a. 24

b. -12

c. -24

d. 12

34 **[Intermediate]** • **Target: The Second Partials Test (Hessian Discriminant)**

For $f(x, y) = 4x^2 - 2xy + y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

a. 12

b. 16

c. 8

d. -12

35 **[Intermediate]** • **Target: Classifying Extrema and Inconclusive Cases**

For $f(x, y) = 3x^2 - xy + 3y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

a. 36

b. 33

c. 35

d. -35

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36 **[Intermediate]** • **Target: Lagrange Multipliers for Constrained Optimization**

For $f(x, y) = 2x^2 - xy + 4y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

a. 32

b. 31

c. 29

d. -31

37 **[Intermediate]** • **Target: Finding Critical Points**

For $f(x, y) = 2x^2 - xy + 2y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

a. 16

b. 13

c. 15

d. -15

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: The Second Partials Test (Hessian Discriminant)** • **Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives**

For $f(x, y) = 3x^2 - 3xy + 3y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

a. 36

b. 27

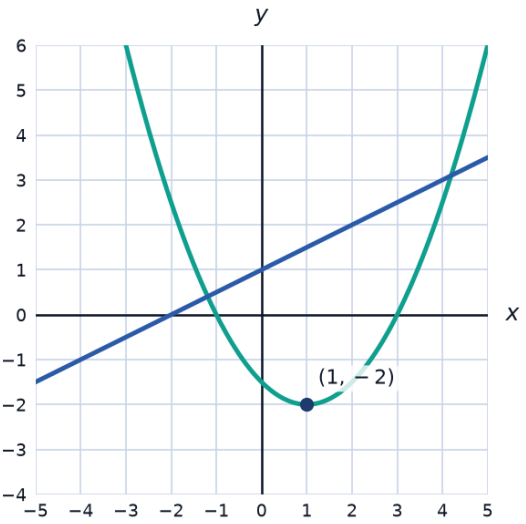
c. 21

d. -27

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39 [Challenging] • Target: Classifying Extrema and Inconclusive Cases • Cross-Lesson: 4.01 Double Integrals

For $f(x, y) = 3x^2 - 2xy + 2y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).



a. 24

b. 16

c. -20

d. 20

40 [Challenging] • Target: Lagrange Multipliers for Constrained Optimization • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

For $f(x, y) = 2x^2 - 2xy + 4y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

a. 32

b. 24

c. 28

d. -28

41 [Challenging] • Target: Finding Critical Points • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

For $f(x, y) = x^2 - 4xy + 2y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

a. 8

b. -8

c. -16

d. -7

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42 [Challenging] • Target: The Second Partials Test (Hessian Discriminant) • Cross-Lesson: 4.01 Double Integrals

For $f(x, y) = 4x^2 - 2xy + 2y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

a. 32

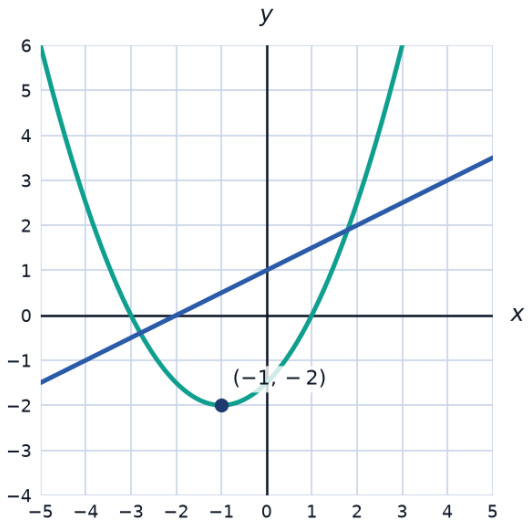
b. 28

c. 24

d. -28

43 [Challenging] • Target: Classifying Extrema and Inconclusive Cases • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

For $f(x, y) = 4x^2 - 5xy + 3y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).



a. 48

b. 13

c. -23

d. 23

44 [Challenging] • Target: Lagrange Multipliers for Constrained Optimization • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

For $f(x, y) = 4x^2 - 2xy + 4y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

a. 64

b. 56

c. 60

d. -60

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45 [Challenging] • Target: Finding Critical Points • Cross-Lesson: 4.01 Double Integrals

For $f(x, y) = 4x^2 - 4xy + 4y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

a. 64

b. 40

c. -48

d. 48

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